Spring Oscillation: A Worked Example

SYNOPSIS

One problem, worked all the way through. Notes 4 and 5 built the theory of simple harmonic motion; this is what it looks like when you are handed two measurements and asked for a graph and a force. Nothing new is introduced — every tool used here is already in those two notes.

BEFORE YOU START — PREDICT

The block in the problem below is oscillating. At which moment is the **force on it largest**: when it is whipping through the middle at top speed, or when it has stopped dead at the far end of its travel? Write down your answer and one sentence of reason. Part 2 settles it.

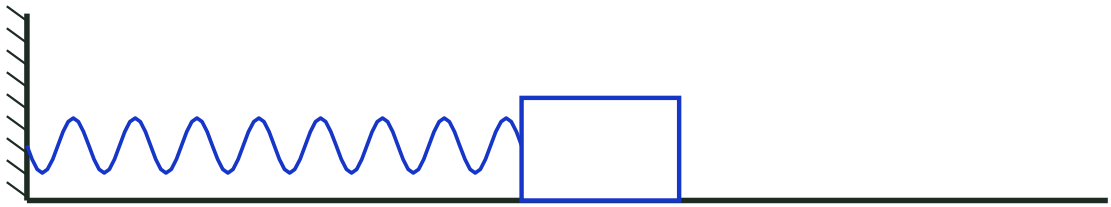
PERFORMANCE EXPECTATIONS

1. **Produce** $x(t)$ for a mass-spring system from two measurements, and sketch it with both axes scaled.
2. **Compute** ω, f and T from k and m, and check the unit of each.
3. **Obtain** the maximum acceleration and the maximum force from the amplitude and ω.
4. **Say** where in the cycle the speed, the acceleration and the force each reach a maximum, and where each is zero.

THE PROBLEM

At $t = 0$, a 500 g block oscillating on a spring is observed moving to the right at $x = 15$ cm. It reaches a maximum displacement of 25 cm at $t = 0.3$ s.

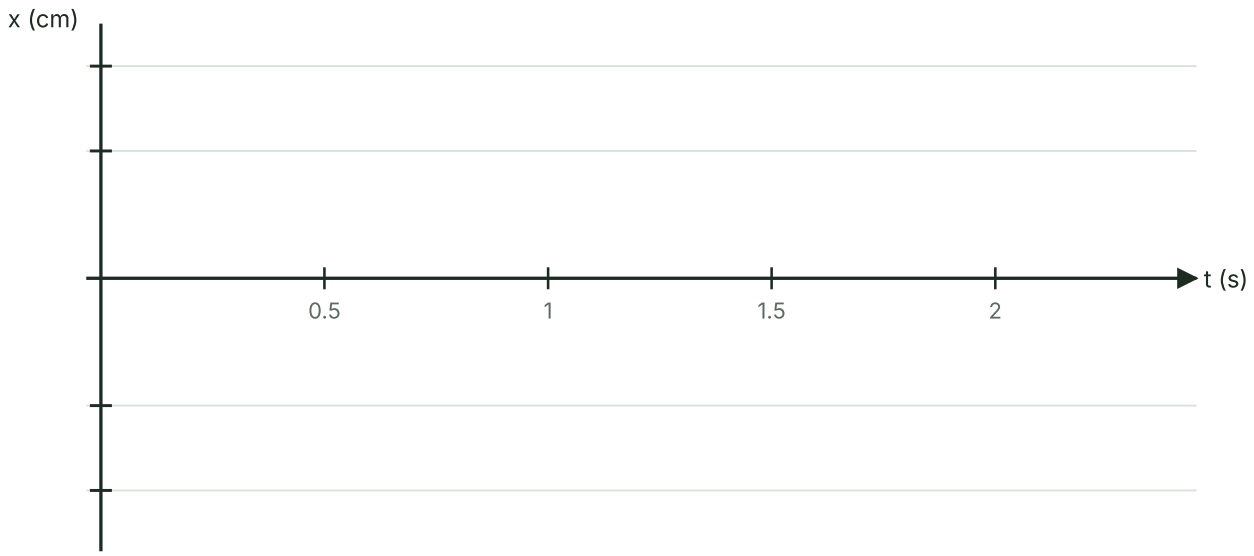
1. Draw a position–time graph for one cycle.
2. What is the maximum force on the block, and when does this occur?



The two things you are given: where it is at $t = 0$, and which way it is going. **The direction is not decoration** — it is what fixes the phase, and it is the only thing in the problem that does.

Part 1 — the graph

1. Write the general form of the position of a simple harmonic oscillator. Which of its three constants does the problem hand you straight away?
2. Use the position at $t = 0$ to find the phase constant φ. There are two angles with the right cosine — what in the problem tells you which one?
3. Use the fact that the maximum displacement happens at $t = 0.3$ s to find ω.
4. Turn ω into f and T, and sketch one full cycle starting from $t = 0$.



Draw one full cycle here. Mark where it starts, where it peaks, and how long the cycle takes.

Part 2 — the maximum force

5. You have A and ω. Write the acceleration of a simple harmonic oscillator and read off its largest value. Mind the units.
6. Turn that into a force, and say **when** it happens — both the first time and every time after.

THE PUNCHLINE

The force is largest exactly where the block is **slowest**, and zero exactly where it is **fastest**. That is not a quirk of this problem — it is $F = -kx$ read out loud. The spring pushes hardest when it is stretched furthest, and at that instant the block has stopped and is about to come back.

Looking ahead

That is the whole of the single-oscillator story: a restoring force proportional to displacement, a sine or cosine in time, and a frequency set by the system rather than by how hard you started it. **Note 7** lets the disturbance leave home — instead of one object going up and down forever in one place, the oscillation is handed from each bit of a medium to the next, and travels.

Where this sits. Everything used here came from Notes 4 and 5: the form of $x(t)$, the fact that ω belongs to the spring and the mass, and $a = -A\omega^2 \cos(\omega t+\varphi)$. The only new skill is reading a phase constant out of a single measurement and a direction.

BACK TO YOUR PREDICTION

Largest at the far end, where it has stopped. If you said "at top speed in the middle", you were thinking of kinetic energy, or of momentum — both of which really are largest there. Force is not either of those. Force follows displacement, and in the middle the displacement, and so the force, is zero.